The Diophantine equation $x^2 = n!! \cdot \mathrm{lcm}(1, \ldots, n)$: the complete solution set, formalized and machine-checked in Lean 4

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Abstract

We determine the complete solution set of $x^2 = n!! \cdot \mathrm{lcm}(1, \ldots, n)$ over the natural numbers: the product is a perfect square if and only if $n \in \{0, 1, 2, 5, 7\}$, with square roots $x = 1, 1, 2, 30, 210$. This is the second of the two equations that Novakovi'c (arXiv:2601.16757) states in his own text as open, and the answer given here is strictly stronger than the finiteness he asks for. The result, an iff-characterization together with its parity criterion and witness layer, is formalized in Lean 4 (12 theorems, 333 lines) and machine-checked: every proof compiles with no `sorry`, and `#print axioms` reports only {propext, Quot.sound, Classical.choice}. An exclusivity search across OEIS value columns, arXiv metadata, seven neighbouring full texts, the public Lean ecosystem and mathlib found no occupying record for the statement itself. This note is a priority claim of record, not a full paper.

1 Introduction

We give the complete solution set of the Diophantine equation $x^2 = n!! \cdot \mathrm{lcm}(1, \ldots, n)$, where $n!!$ is the double factorial and $\mathrm{lcm}(1, \ldots, n)$ is the least common multiple of the integers from 1 to n. The equation is the second of the pair that Novakovi'c (arXiv:2601.16757) lists as open in his own text; a companion result settled the first equation $x^2 = n! \cdot n\#$. This note is a priority claim of record for the present result: the narrative paper follows separately. The formalization was checked by the Lean 4 kernel, with no `sorry` and an axiom set limited to the three standard proof-relevant axioms.

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2 Statement

Theorem 1. *For every natural number n, $n!! \cdot \operatorname{lcm}(1, \ldots, n)$ is a perfect square if and only if $n = 0$ or $n = 1$ or $n = 2$ or $n = 5$ or $n = 7$.*

Theorem 2. *For every natural number n, $n!! \cdot \operatorname{lcm}(1, \ldots, n)$ is a perfect square if and only if, for every prime p, the total exponent $\nu_p(n!!) + \lfloor \log_p n \rfloor$ is even.*

Theorem 3. *The five solution points hold: $1^2 = 0!! \cdot \operatorname{lcm}(1, \ldots, 0)$, $1^2 = 1!! \cdot \operatorname{lcm}(1, \ldots, 1)$, $2^2 = 2!! \cdot \operatorname{lcm}(1, \ldots, 2)$, $30^2 = 5!! \cdot \operatorname{lcm}(1, \ldots, 5)$ and $210^2 = 7!! \cdot \operatorname{lcm}(1, \ldots, 7)$.*

Listing 1: Formal statements (Lean 4, frozen snapshot `9baa672f`).

```
theorem isSquare_dfact_mul_lcmUpto_iff (n : ℕ) :
    IsSquare (n!! * Nat.lcmUpto n) ↔
      ∀ p : ℕ, p.Prime → Even ((n!!).factorization p + p.log n)
    := by
  rw [Nat.isSquare_iff_even_factorization]
  refine forall_congr' fun p => imp_congr_right fun hp => ?_

theorem dfact_lcm_witnesses :
    1 ^ 2 = 0!! * Nat.lcmUpto 0 ∧ 1 ^ 2 = 1!! * Nat.lcmUpto 1 ∧
      2 ^ 2 = 2!! * Nat.lcmUpto 2 ∧ 30 ^ 2 = 5!! * Nat.lcmUpto 5
    ∧
        210 ^ 2 = 7!! * Nat.lcmUpto 7 := by
  decide

theorem dfact_lcm_square_solution_set (n : ℕ) :
    IsSquare (n!! * Nat.lcmUpto n) ↔ n = 0 ∨ n = 1 ∨ n = 2 ∨ n =
     5 ∨ n = 7 := by
```

3 Proof

The complete machine proof is in `proofs/final-dfact-lcm-square.lean` of this package (333 lines, 12 theorems). The key excerpt below shows the opening of the main theorem: the forward direction is by contradiction, splits at $n = 4$, and dispatches the two remaining ranges to the even arm, the odd arm and the witness layer.

Listing 2: Excerpt: the main theorem, forward direction (frozen artifact, sha256 `1107cad4`).

```
theorem dfact_lcm_square_solution_set (n : ℕ) :
    IsSquare (n!! * Nat.lcmUpto n) ↔ n = 0 ∨ n = 1 ∨ n = 2 ∨ n =
     5 ∨ n = 7 := by
  constructor
  · intro h
    by_contra hne
    push_neg at hne
    rcases lt_or_ge n 4 with hlt | hge
```

The proof has three parts. Theorem 2 reduces squareness to a parity condition on prime exponents, using $\mathrm{lcm}(1,\ldots,n)$'s factorization $\nu_p(\mathrm{lcm}(1,\ldots,n)) = \lfloor \log_p n \rfloor$. Even $n \geq 4$ are excluded by Bertrand's postulate: a prime $p \in (n/2, n]$ contributes $\nu_p(n!!) = 0$ and $\lfloor \log_p n \rfloor = 1$, an odd total. Odd $n \geq 33$ are excluded by an interval prime lemma for the interval $(n/4, n/3]$, proved in the accompanying library development from a finite certificate chain and a tail estimate, not as an instance of a published interval theorem, and imported by the final file rather than restated in it: the prime p found there contributes $\nu_p(n!!) = 2$ and $\lfloor \log_p n \rfloor = 1$, again an odd total. The remaining band $9 \leq n \leq 32$ and the small cases $n = 3$ are handled by six explicit non-square witnesses with named odd-exponent primes. Theorem 3 supplies the five solution points, which gives the converse direction.

4 Verification evidence

- Toolchain: Lean 4 v4.34.0, mathlib4 v4.34.0 (rev `5ed29652`).
- Statement integrity: the frozen statement file has sha256 `9baa672fa40e74626369e4803d66bedd9b`
  the final proof file has sha256 `1107cad44affa0bc84bb4a285c0381d6e6a0afd15c08d935cbda0cee`
- Kernel check: strict-mode compilation of the final file exits 0 with no `sorry` (the only diagnostic is one deprecation warning on `push_neg`). `#print axioms` output for the twelve theorems:

```
'DFactLcmSquare.isSquare_dfact_mul_lcmUpto_iff' depends on axioms: [propext, Cla
'DFactLcmSquare.dfact_lcm_witnesses' depends on axioms: [propext, Classical.choi
'DFactLcmSquare.dfact_lcm_square_solution_set' depends on axioms: [propext, Clas
'DFactLcmSquare.exists_sq_dfact_lcm_iff' depends on axioms: [propext, Classical.
```

- Audit trail: the frozen statement's kernel well-definedness record, the strict compilation log, the axiom print, the statement diff (clean, declaration by declaration), the degenerate-probe outputs (the statement is load-bearing: perturbing the solution set turns the proof red) and the independent final audit's verdict are in `audit/` of this package.
- The deposited `.lean` files are copies in which comments carrying pre-publication workflow narrative were blanked; the blanking never touches code, and per-file byte equality of the comment-free code view is asserted by the generating script (record: `audit/comment-elision-record.md`).

5 Novelty evidence

The exclusivity search's four-state adjudication is **no occupying record found**, with the statement layer clean at every channel that reaches it.

- **OEIS value columns.** The solution-n column, the solution-root column $x = 1, 1, 2, 30, 210$, the object value column $a(n) = n!!\cdot\mathrm{lcm}(1, \ldots, n)$ (four prefix variants, plus alternating-parity sub-columns and drop-prefix cuts in the review pass), the square value column and the squarefree-part columns were searched, together with keyword layers. The object and solution-root columns are **null** across all prefix variants. The residual hits are unrelated items, or A061742 $= (p_n\#)^2$, which is entered precisely because $x = n\#$ and is not a registration of this equation.
- **arXiv metadata.** Fourteen load-bearing query forms were run, with the positive control `all:"double factorial"` $= 34$ hits confirming the channel was live: no same form at the metadata layer.
- **Neighbouring full texts.** Seven items were read in full (2601.16757, 2510.24312, 2602.09687, 2602.23838, 2604.09730, 2610.01899, 2606.23485), together with all 17 references of the anchor: zero coverage of this equation.
- **Citation axis.** Semantic Scholar and OpenAlex, two independent channels, on the anchor: `cited_by_count` $= 0$ on both, i.e. the anchor's open case has not been closed by anyone citing it.
- **Public Lean ecosystem.** The whole public-conjectures repository at content level: `doubleFactorial` occurs in exactly one file (A001818, a different object); the statement layer has zero registration.

- **mathlib.** Ripgrep with the positive control `Nat.Prime` = 21 files: `doubleFactorial` occurs in 4 files, none in a number-theoretic square context; `lcmUpto` in one; the two objects never co-occur.

Two boundaries are recorded explicitly rather than left implicit. First, the $\mathrm{lcm}(1,\ldots,n)$ side of the object is occupied at the sequence level: OEIS A056839 registers the squarefree part of $\mathrm{lcm}(1,\ldots,n)$. That entry does not mention the double factorial anywhere, and it registers a sequence, not the characterizing solution set of this equation. Second, the review layer records that Crossref is a relevance-ranked channel and that the OpenAlex title search missed the anchor by indexing status; neither is used as a load-bearing zero hit. The full query record, including every zero hit, is in `audit/novelty-record.md` of this package.

6 Scope and limitations

The graded conclusion for this result is *complete proof*: the twelve theorems compile strictly with zero `sorry` and an axiom set limited to {propext, Quot.sound, Classical.choice}, and the statements match the frozen snapshot declaration by declaration. Three boundaries are stated rather than implied. The interval prime lemma for $(n/4, n/3]$ used in the odd arm is proved in the accompanying library development as a finite certificate chain plus an Erdős-style tail argument; this note does not claim it as an instance of any published interval theorem, and it is imported rather than shipped, so a kernel compile of the deposited final file needs those library modules beside it (their identity is recorded in the audit record of this package). The observation that the solution roots are $x = n\#$ is a numerical observation recorded alongside the theorem, not a separate formalized statement. This note contains no literature survey and makes no claim about related equations: the companion equation $x^2 = n! \cdot n\#$ was settled separately, and the intersection $\{5, 7\}$ of this solution set with the Brocard solutions $\{4, 5, 7\}$ is a numerical coincidence with no implication in either direction.

Data availability

The Lean sources, the compile and axiom records and the exclusivity-search record are packaged on Zenodo (DOI: `10.5281/zenodo.23309733`). The note is released under CC BY 4.0 and the code under MIT.

Use of generative AI

The formalization and proof reported in this note were developed with AI assistance (large language model: `kimi-k3`) under human direction and review. All statements and proofs reported here were checked by the Lean 4 kernel (Lean v4.34.0, mathlib v4.34.0): every proof compiles with no `sorry`, and `#print axioms` reports only {propext, Quot.sound, Classical.choice}. The author reviewed all content and takes full responsibility for it.

Author contributions

The author selected the problem, verified the semantics of the formal statements against the informal ones, and reviewed the kernel evidence and the exclusivity record before signing off this deposit.

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